Instagram Engagement Prediction Model

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Problem Statement

- Social media engagement is hard to predict
- Businesses rely on engagement for growth
- Need data-driven strategy

Project Objectives

- Predict engagement rate
- Identify key drivers of engagement
- Compare machine learning models

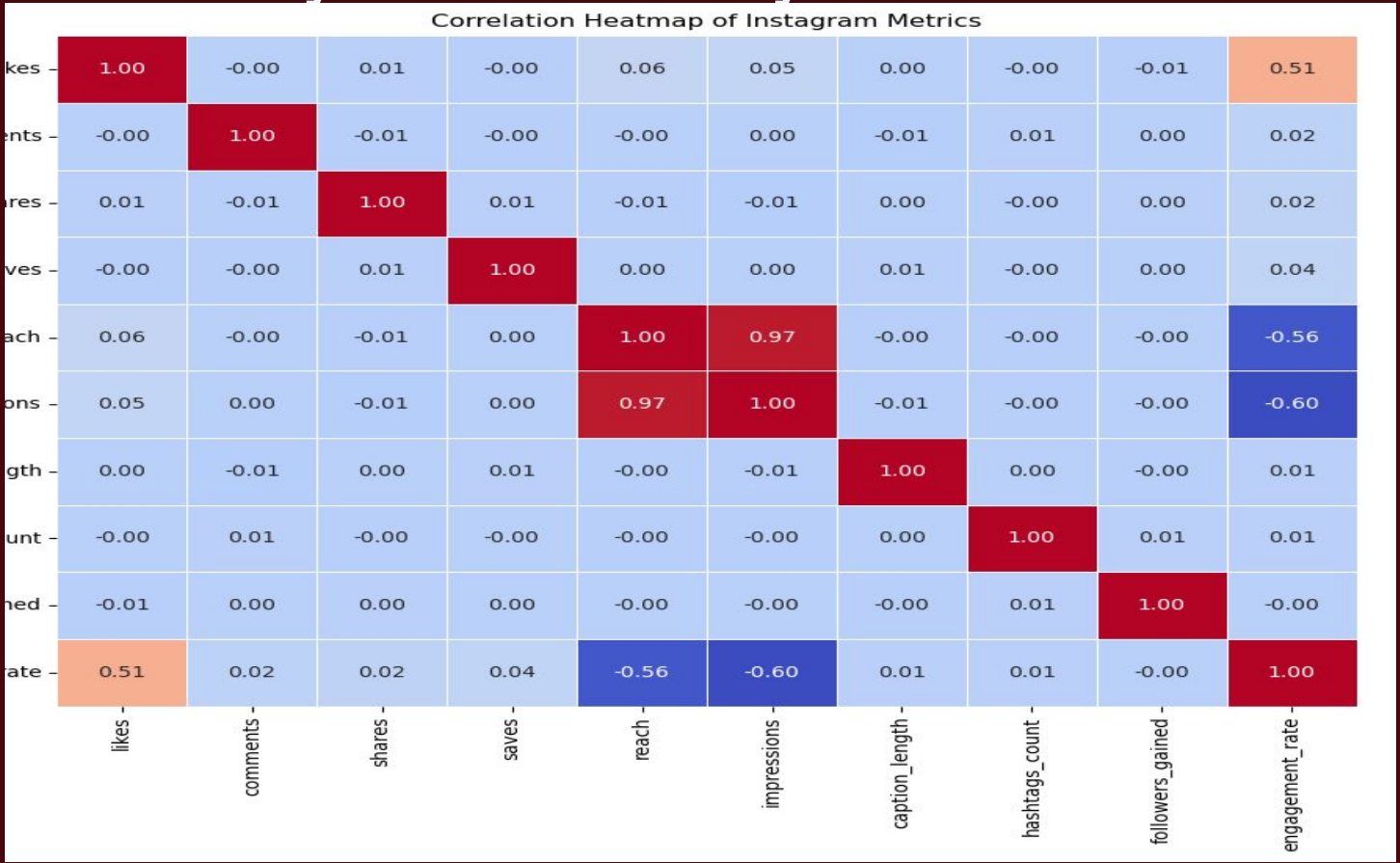
Dataset Overview

- Instagram analytics dataset
- 10,000 rows
- Features: Likes, comments, shares, saves, impression

Data Cleaning and Preprocessing

- Removes/Handled missing values
- Feature engineering (engagement rate)
- Converted data types

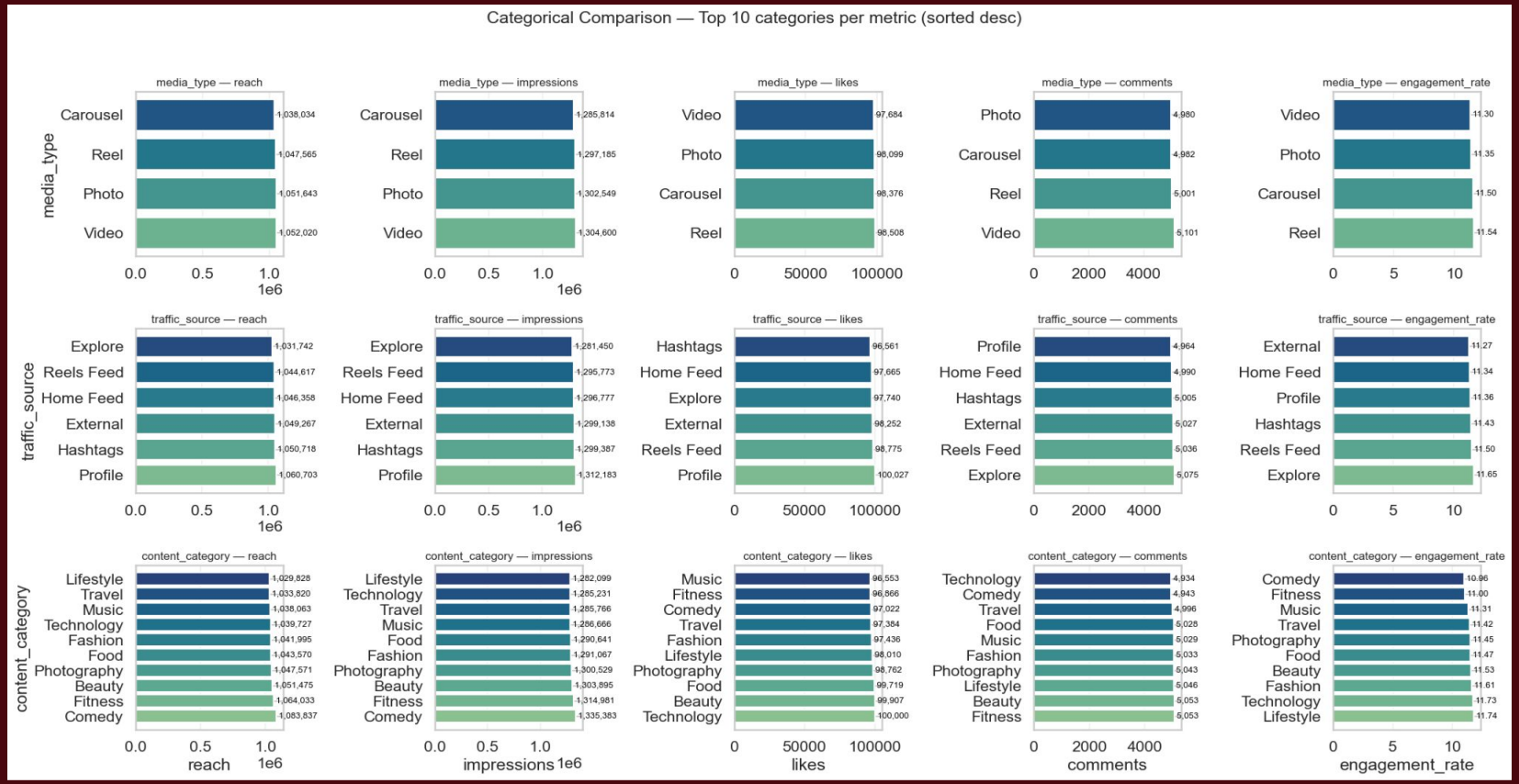
Exploratory Data Analysis



Exploratory Data Analysis Cont’d

- Correlation Heatmap of Instagram Metrics
- Strong positive correlation: reach and impression (0.97)
- Engagement rate positively correlated with likes (0.51)
- Engagement rate negatively correlated with reach (-0.56) and impressions (-0.60)

Category Performance



Modeling Approach

- Linear Regression
- Random Forest
- KNN

Model Evaluation Metrics

- RMSE
- MAE
- R^2

Model Performance Comparison

Models	RMSE	MAE	R2
Random Forest	0.7439	0.2983	0.9946
Linear Regression	5.9085	3.6592	0.6623
KNN Regressor	6.8589	3.6267	0.5450

Reality Check /Limitations

- Possible data leakage
- Engagement formula overlap
- Overfitting Risk

Final Model Selection

- Random Forest selected
- Best performance but with caution
- Overfitting Risk

Key Insights

- Shareable content drives engagement
- Saves and shares are critical

Business Impact

- Optimize content strategy
- Focus on high-engagement features

Recommendations

- Optimize content strategy
- Focus on high-engagement features

Future Work

- Add more features
- Try deep learning models
- Improve data quality

Conclusion

- Model predicts engagement effectively
- Results require careful interpretation